Engine Model(s): PT6A

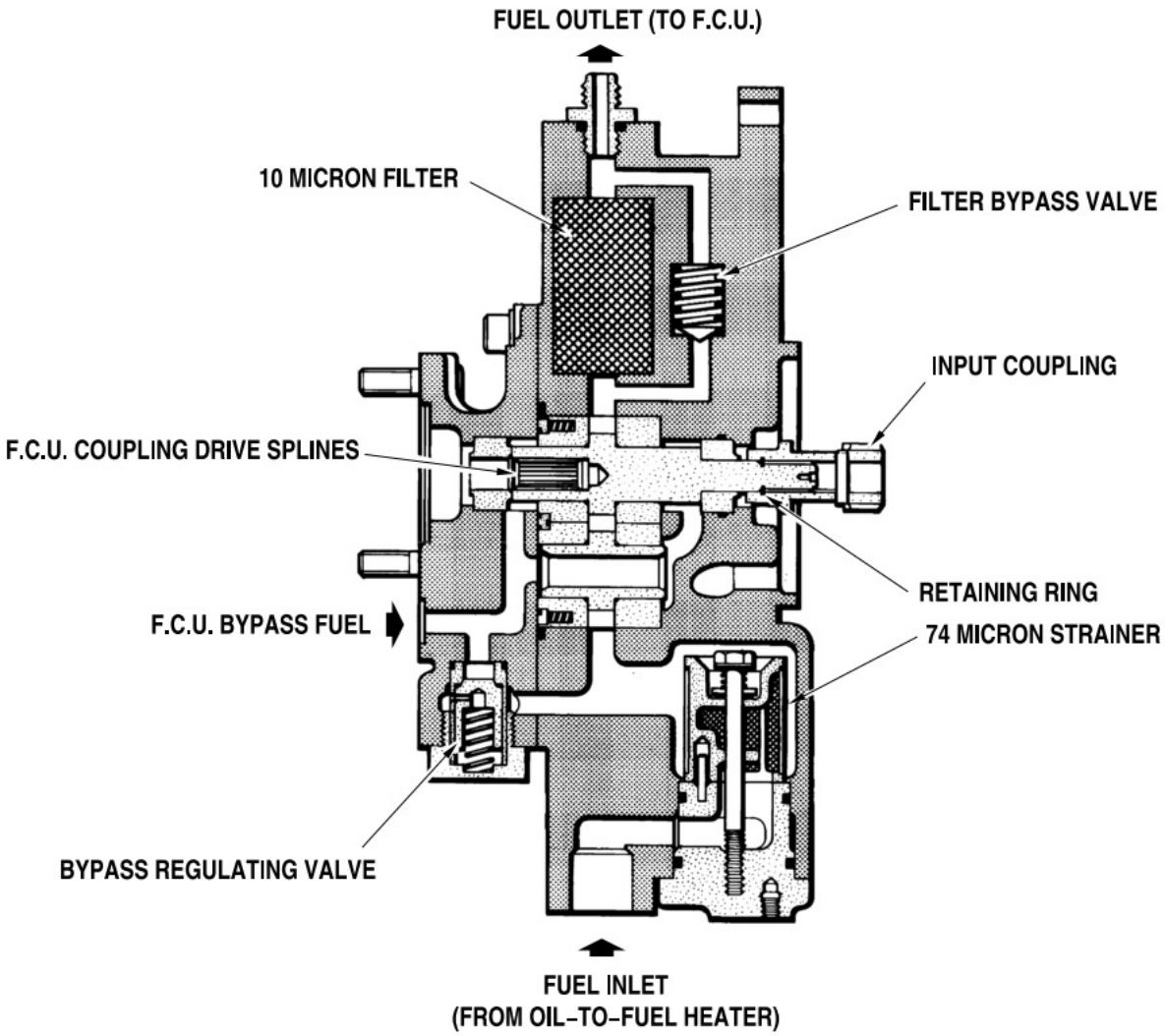
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FUEL PUMP - DESCRIPTION AND OPERATION

1. Description Operation (Ref. Fig. 1)

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Figure 1 Fuel Pump



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The engine-driven fuel pump is a positive displacement gear-type pump, mounted on a pad at the 2 o'clock position on the rear face of the accessory gearbox. The pump incorporates spring and pressure loaded bushings and is shaft-driven off the accessory gearbox. A splined coupling is used to transmit the geartrain drive to the pump gears; the coupling splines are lubricated by oil mist from the gearbox via a hole in the associated gearshaft. Another coupling, on the other side of the pump gears, is used to transmit a speed signal to the fuel control unit (FCU).

Fuel from the oil-to-fuel heater enters the fuel pump chamber through a 74-micron (200-mesh) inlet screen. The inlet screen is spring-loaded and, should it become blocked, the increase in differential pressure overcomes the spring and allows unfiltered fuel to flow into the pump chamber. The pump gears increase the fuel pressure and deliver it to the FCU via a 10-micron filter in the pump outlet and an external transfer hose assembly. Seepage fuel drain connection points are provided to conduct fuel to an airframe collection point. A bypass valve, and cored passages in the pump body, enable unfiltered high pressure fuel to flow to the FCU in the event of the outlet filter becoming blocked. An internal passage originating at the mating face of the FCU returns bypass fuel from the FCU metering valve directly into the pump gears. A pressure regulating valve in the bypass line serves to pressurize the bearings and direct excess bypass fuel to the pump fuel inlet, downstream of the inlet screen.

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